

Déjà Vu: An Illusion of Prediction

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**Abstract**

Déjà vu is beginning to be scientifically understood as a memory phenomenon. Despite recent scientific advances, a remaining puzzle is the purported association between déjà vu and feelings of premonition. Building on research showing that déjà vu can be driven by an unrecalled memory of a past experience that relates to the current situation, we sought evidence of memory-based predictive ability during déjà vu states. Déjà vu did not lead to above-chance ability to predict the next turn in a navigational path resembling a previously experienced but unrecalled path (although such resemblance increased reports of déjà vu). However, déjà vu states were accompanied by increased feelings of knowing the direction of the next turn. The results suggest that feelings of premonition during déjà vu occur and can be illusory. Metacognitive bias brought on by the state itself may explain the peculiar association between déjà vu and the feeling of premonition.

Keywords

déjà vu, cognitive bias, bias, prediction, familiarity, open data, open materials, preregistered

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Déjà vu—the striking feeling of having experienced something before despite knowing otherwise (Brown, 2003)—is beginning to be scientifically understood as a memory phenomenon (Brown & Marsh, 2008, 2009, 2010; Cleary, 2008; Cleary et al., 2012; Cleary, Ryals, & Nomi, 2009). It can be elicited by an unrecalled memory for a past experience that relates to the current situation (Cleary et al., 2012) and can also accompany seizure activity in medial temporal regions of the brain related to memory (Bowles et al., 2007). Despite this advance in understanding the basis of déjà vu, a remaining scientific puzzle is the purported association between déjà vu and feelings of premonition, or knowing what will happen next. People often report such an association from personal experience in surveys (Brown, 2004). Case-study research on déjà vu elicited by cortical stimulation also uncovered this association: Cortical-stimulation-induced déjà vu in a patient was accompanied by a feeling of knowing exactly what would happen next (Mullan & Penfield, 1959, as cited in Brown, 2004). In the present study, we aimed to experimentally investigate the suggested association between déjà vu and feelings of premonition.

Our theoretical rationale was as follows. It has been argued that memory's primary adaptive function is to

allow prediction of the future more so than to enable recollection of the past (Schacter, Addis, & Buckner, 2007; Szpunar, Spreng, & Schacter, 2014). As a memory-driven signal of familiarity during recall failure (Cleary et al., 2012), déjà vu might plausibly be associated with thinking about the future. Specifically, an unrecalled memory for how a previously experienced, dynamic event unfolded could lead not only to a sense of déjà vu for a seemingly new but related current situation (as in Cleary et al., 2012), but also to a sense of what will happen next as the event continues to unfold. In short, if memory in general is connected to thinking about the future (e.g., Schacter et al., 2007; Szpunar et al., 2014), then it follows logically that subjective states of memory during recall failure, too, could be linked to thinking about the future. If so, this could potentially explain the purported link between déjà vu and feelings of premonition.

To elaborate, if one had previously experienced a particular event in the past, then one would be in a

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position to potentially recollect how that event unfolded over time. Even if recollection failed, the feeling of familiarity might lead to the tingling sense that recollection is imminent, similar to a tip-of-the-tongue (TOT) state (Brown, 2012), giving rise to the feeling that one knows what will happen next (i.e., that the knowledge is available, if not at this moment accessible). With regard to TOT states, there is evidence that partial information about a sought-after word is sometimes accessible (Brown, Burrows, & Croft Cadarao, 2013). Indeed, it has been argued that it is the accessibility of such partial information that gives rise to the TOT state (Brown, 2012). Similarly, it might be that accurate retrieval of information about “what happens next” may contribute to *déjà vu*. On the other hand, it might be that the familiarity of an event that is similar to a non-recalled event merely gives rise to the illusion that one knows what happens next. The current research was designed to explore these possibilities.

To examine whether *déjà vu* and feelings about the future are connected, we created a dynamic variant of a previously used paradigm for examining reports of *déjà vu* in the laboratory. The previous paradigm used virtual reality to demonstrate that an identical spatial layout to a previously viewed but unrecalled scene increases the likelihood of reported *déjà vu* for an otherwise novel scene (Cleary et al., 2012). Although that study supports the hypothesis that *déjà vu* can be memory driven (driven by an unrecalled memory for something relating to the current situation), the paradigm lacked the dynamicity of events needed to examine the relationship between *déjà vu* and the sense of what might happen next as a situation unfolds. To give the previously used scenes dynamicity, we created video of navigation through each scene from a first-person perspective. As in the prior virtual reality study, each test scene was novel, but half had the same spatial arrangement of elements as a previously viewed scene (Fig. 1).

In the present study, the navigational path taken through each test scene was identical to that taken in the configurally identical scene to which it mapped, but movement through the scene stopped before a critical turn, which was either left or right. On the basis of prior research (Cleary et al., 2012), we hypothesized that configural resemblance to an unrecalled but earlier-viewed scene should increase the likelihood of reporting *déjà vu* for that scene. Our interest was in (a) whether reported *déjà vu* would be associated with an above-chance ability to predict in which direction the navigation should proceed (left vs. right) on the basis of an unrecalled memory for the navigational path taken in a configurally similar scene viewed earlier and (b) whether reported *déjà vu* would be associated with

an increased sense of *thinking* that one knows in which direction the navigation should proceed next.

Method

Participants

Seventy-four Colorado State University students participated in Experiment 1 in exchange for course credit (an additional participant was excluded for typing random keys as responses). In Experiments 2 and 3, 103 and 121 participated, respectively. We aimed for a large sample size relative to past research, plus replication, because participants might, in fact, have no ability to predict the direction of the turn during retrieval failure at above-chance levels (thus, null prediction effects might be shown and would be critical to the overall pattern obtained, warranting a fairly large sample size). Experiment 3 was preregistered at the Open Science Framework for 120 participants. The number was based on a conservative power analysis (with power set to .90, an effect size, or d , of .30, and an α of .05), which indicated that our required sample size was 119. We rounded up to 120 for our preregistration to achieve an even number. We then targeted 120 total participants through our weekly sign-up process over the course of many weeks and captured 1 additional participant through that estimation process, who we included (arriving at 121).

Participants partook in each experiment individually on a Dell computer using E-Prime software (Version 2.0; Schneider, Eschman, & Zuccolotto, 2012). Participants in Experiments 1 and 2 each participated alone in a private booth with sound driven through speakers. Participants in Experiment 3 had an experimenter in the room with them in an effort to encourage attention to the task at hand. Some of Experiment 3's participants were run in a private booth with an experimenter sitting in the booth with them (sound was driven through speakers in this case); others were run in a room of eight computers with up to 8 participants at once and an experimenter present in the room (sound was driven through headphones for these participants, and privacy screens shielded monitors from being viewable by other participants).

Materials

Stimuli were videos of navigation through the 64 pairs of configurally similar scenes that were used in a previous virtual reality study of *déjà vu* (Cleary et al., 2012). These existing scene pairs were identical in their spatial layout of elements (see Figs. 1a and 1b and Figs. 1c and 1d, respectively). The navigational videos were created using the video mechanism of The Sims game

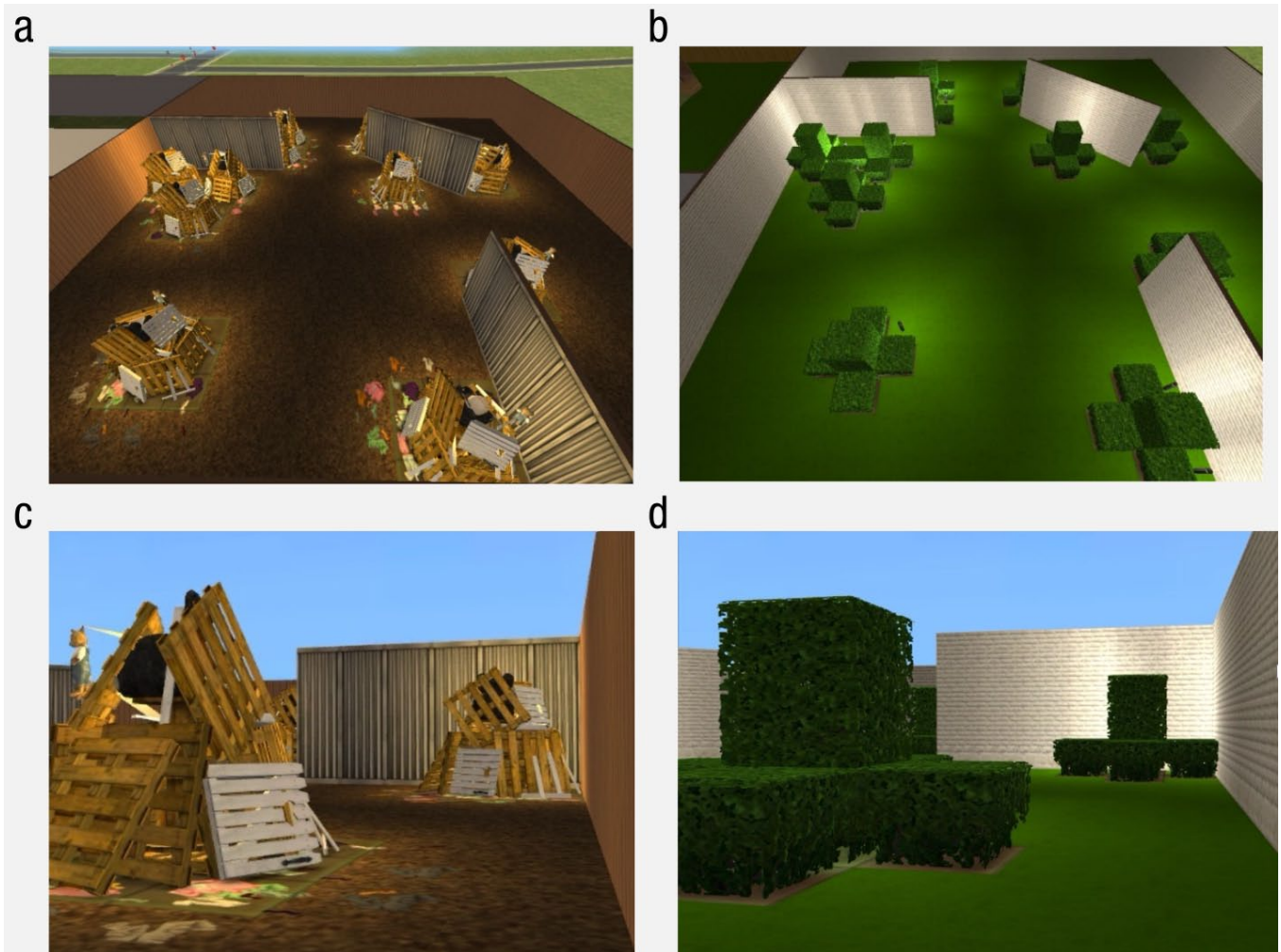


Fig. 1. Stimulus examples. The top row shows a bird's-eye view of (a) a junkyard scene and (b) a hedge garden test scene identically mapped to it in configuration. Such configural mapping contributes to déjà vu reports when retrieval of the prior mapping scene fails (Cleary et al., 2012). The bottom row shows the participant's perspective of video navigation through (c) the junkyard scene and (d) the configurally mapped hedge garden scene. At encoding, participants viewed videos depicting first-person navigation through scenes such as the one depicted in (c). At test, participants viewed videos of navigation through scenes not viewed earlier, such as the hedge garden depicted in (d). For a given test scene, such as the one shown in (d), the same navigational path was used in the video as in the earlier-viewed configurally similar scene, but the movement stopped short of a critical turn (left or right) that occurred in the earlier-viewed navigation. On stopping short of the next turn, participants judged the direction of the next turn. Our interest was in when retrieval of the configurally mapping scene failed: Could participants still predict the direction of the next turn on the basis of that unretrieved memory? Would such an ability be associated with reporting déjà vu? Is déjà vu associated with an increased feeling of knowing the next turn?

platform that was used to create virtual reality environments in previous work (Cleary et al., 2012). Videos of first-person navigation through each of the 64 pairs of configurally similar scenes were created such that for a given pair of configurally similar scenes (see Figs. 1c and 1d), the same navigational path of exploration through the scene was used, with the exception that in what was to be the configurally similar test scene (e.g., Fig. 1d), the navigational movement stopped short of a critical turn that had occurred in its corresponding configurally identical scene (e.g., Fig. 1c). This was so that participants could make a judgment about whether

the critical turn in the test scene should be right or left. Correspondence of test scenes to studied versus non-studied scenes, as well as to left versus right critical turns, was counterbalanced across participants. This counterbalancing was achieved by creating four versions of the experiment altogether. Two versions of the video for each primary scene (such as that shown in Fig. 1c) were created: one in which the critical turn was left and one in which the critical turn was right. Two versions of each of these sets were created to also balance which test scenes corresponded to studied versus non-studied scenes. Finally, for each potential study scene,

an audio file was created to play while the video was being shown. Each audio file was a .wav recording of a female voice stating the name of the scene and then repeating that name (e.g., "This is a junkyard. Junkyard.").

Procedure

Participants received two study-test blocks, each containing a study list of 16 videos of navigation through a scene, followed by a test list of 32 videos of navigation through a novel set of scenes. Though none of the randomly ordered test scenes had appeared at study, half mapped onto a studied scene in element configuration (e.g., Figs. 1a and 1b), and half did not. After the initial instructions were presented, participants pressed a key to begin viewing the first study list of videos. As the video of navigation was being shown, the name of the scene (e.g., "aquarium") was spoken through a set of speakers or headphones as described above. For example, for the aquarium scene, the voice would state "This is an aquarium. Aquarium." Experiment 3, which was intended to increase attention to the navigational path at encoding, prompted participants to type in the direction of the final turn (left or right), followed by a prompt to then type the name of the scene after each video. Once the 16 navigational videos and their scene names had been presented in the initial encoding phase, the instructions for the test were presented.

Before the test began, participants were told that "Déjà vu is the feeling of having been someplace or done something before, without being able to pinpoint why and despite knowing that the current situation is new." They were told that they would see a series of videos and that some would be similar but not identical to videos that they viewed earlier in the study phase and that following each navigational video clip on the test, they would be prompted with several questions pertaining to that test video. The sequence of questions was described in the instructions, and after reading through the instructions, participants could proceed to the test list by pressing any key.

The 32 test videos were presented in a random order. For each test video, the final frozen screenshot of where the video navigation stopped short of a turn remained on the screen as the sequential set of dialog boxes appeared in the center of the screen. First, participants were asked about their inclination to turn right or left where the video left off. They were prompted with, "Without knowing why, you may feel a sense of which way to turn next. Indicate which way to turn. Press L for Left and R for Right." Next, a dialog box asked "Did this scene prompt you to experience Déjà vu? (Y=yes, N=no)." Following that response, participants were prompted with a dialog box that asked, "How familiar

did this scene seem to you on a scale of 0-10? 0 = not at all familiar; 10 = very familiar." Finally, they were asked: "Did this scene remind you of an earlier scene from study? If so, type that earlier scene's name; otherwise, hit ENTER."

In Experiments 2 and 3, an additional question was added to the sequence of prompts on each test trial, and the order of prompting was changed from Experiment 1 to accommodate this. At the end of a given test video clip, participants were first prompted to indicate whether they were experiencing déjà vu for the scene. Then they were prompted to rate their sense of knowing in which direction the camera should turn next from where the video left off (left or right). Hereafter, we refer to this as the *feeling-of-prediction* judgment. For this judgment, participants were specifically asked to give a rating between 0 (*absolutely no sense of knowing in which direction the camera would turn next*) and 10 (*a very strong sense of knowing in which direction the camera would turn next*). Then they were asked to judge whether the turn should be left or right by typing "L" for left or "R" for right. Following this, they were asked to rate how familiar the test scene seemed on a scale of 0 (*not at all unfamiliar*) to 10 (*very familiar*). Finally, they were asked to type in the name of the corresponding configurally similar scene if they could remember one from study that mapped onto the test scene, pressing "Enter" if they could not think of one.

Results

For all *t* tests, we present 95% confidence intervals (CIs) around the means and 95% confidence intervals around effect sizes. For one-sample *t* tests, the effect sizes are Cohen's *ds*. For paired-samples *t* tests, we used d_z with a correction for bias as specified in Viechtbauer (2010). We used the "escalc" method from the *metafor* package (Viechtbauer, 2010) to calculate the confidence intervals around d_z .¹

As a manipulation check, among test scenes that mapped onto studied scenes, when recall of the previously viewed scene succeeded (e.g., a participant recalled the junkyard scene in response to viewing the hedge garden scene depicted in Fig. 1), there was above-chance (chance = .50) prediction of whether the next turn in the test scene should be left or right in all three experiments (note that the degrees of freedom in each reflect the fact that not all participants exhibited successful recall). Performance was above chance in Experiment 1 ($M = .55$, 95% CI = [.51, .60], $t(72) = 2.34$, $p = .022$, $d = 0.27$, 95% CI = [.01, .43], and Experiment 2 ($M = .57$, 95% CI = [.52, .62], $t(91) = 3.15$, $p = .002$, $d = 0.33$, 95% CI = [.11, .55]. Though only directional

and not significant in a cross-experiment comparison, it was highest in Experiment 3, where additional focus at encoding was required ($M = .61$, 95% CI = [.58, .65], $t(119) = 6.02$, $p < .001$, $d = 0.55$, 95% CI = [.35, .75]. Presumably, in all three experiments, participants' recollection of the studied video's navigational path through the scene enabled this above-chance prediction when recall succeeded.

As in prior research (Cleary et al., 2012), when recall of the previously viewed configurally similar scene failed, there was still evidence of memory for the earlier encounter despite the retrieval failure. Specifically, the mapping of a test scene's spatial layout to the layout of an unrecalled earlier viewed scene (Fig. 1) increased the probability of reporting déjà vu. In Experiment 1, the probability of reporting déjà vu during recall failure was greater for scenes that spatially mapped onto an unrecalled earlier-viewed scene ($M = .32$, 95% CI = [.26, .38]) than for scenes that did not ($M = .29$, 95% CI = [.24, .33]), $t(73) = 2.12$, $p = .038$, $d_z = 0.23$, 95% CI = [0.014, 0.477]. Again in Experiment 2, the probability of reporting déjà vu was greater for scenes that spatially mapped onto earlier-viewed but unrecalled scenes ($M = .33$, 95% CI = [.29, .37]) than for scenes that did not ($M = .27$, 95% CI = [.24, .30]), $t(102) = 4.85$, $p < .001$, $d_z = 0.45$, 95% CI = [0.25, 0.65]. Finally, this pattern emerged in Experiment 3 also, with a greater probability of reporting déjà vu for scenes that mapped onto earlier-viewed but unrecalled scenes ($M = .30$, 95% CI = [.26, .34]) than for scenes that did not ($M = .24$, 95% CI = [.21, .27]), $t(120) = 5.25$, $p < .001$, $d_z = 0.50$, 95% CI = [0.31, 0.69]. In Experiment 1, familiarity ratings during retrieval failure were higher for configurally similar ($M = 2.85$, 95% CI = [2.50, 3.22]) than dissimilar scenes ($M = 2.53$, 95% CI = [2.22, 2.86]), $t(73) = 3.52$, $p = .001$, $d_z = 0.41$, 95% CI = [0.17, 0.64]. The same pattern held in Experiment 2, with higher familiarity ratings for configurally similar ($M = 2.83$, 95% CI = [2.50, 3.12]) than dissimilar scenes ($M = 2.47$, 95% CI = [2.19, 2.75]), $t(102) = 4.28$, $p < .001$, $d_z = 0.43$, 95% CI = [0.23, 0.63], as well as in Experiment 3 for configurally similar ($M = 2.67$, 95% CI = [2.37, 2.95]) than for dissimilar ($M = 2.22$, 95% CI = [1.99, 2.45]) test scenes, $t(120) = 5.60$, $p < .001$, $d_z = 0.51$, 95% CI = [0.32, 0.70]. This overall pattern across the three experiments replicates an association between the level of familiarity brought on by configural resemblance and reports of déjà vu during recall failure (Cleary et al., 2009; Cleary et al., 2012).

Despite the detectable familiarity and memory-driven déjà vu for the test scenes in cases of recall failure, there was no above-chance predictive ability in any of these individual cases regarding in which direction the next turn should be. The probability of correctly choosing a

direction was no greater than chance (.50) in Experiment 1 ($M = .52$, 95% CI = [.49, .55]), $t(73) = 1.12$, $p = .27$, $d = .13$, 95% CI = [−.27, .53]; Experiment 2 ($M = .52$, 95% CI = [.50, .54]), $t(102) = 1.55$, $p = .124$, $d = 0.15$, 95% CI = [−0.11, 0.41]; or Experiment 3 ($M = .51$, 95% CI = [.50, .54]), $t(120) = 1.35$, $p = .179$, $d = 0.12$, 95% CI = [−0.14, 0.38].

To assess the likelihood that our data favored the null hypothesis (that the mean was not different from .50) in each experiment, we performed Bayes factor analyses (e.g., Jarosz & Wiley, 2014). Given that our particular experiments are novel and have never before been done, we followed the recommendations of Rouder, Speckman, Sun, Morey and Iverson (2009) for one-sample t tests. Rouder et al. recommend the Jeffreys-Zellner-Siow (JZS) prior in cases where there is no preexisting prior information or the researchers wish not to commit to any assumptions about priors, and specifically recommend a JZS Bayes factor as a default for performing Bayesian t tests (p. 231). Using their freely available web resource for performing the JZS Bayes factor analysis (<http://pcl.missouri.edu/bf-one-sample>)—which used Version 0.9.8 of the BayesFactor package (R Version 3.3.2) on i386-redhat-linux-gnu to compute the Bayes factor—we obtained a JZS Bayes factor (BF_{01}) of 4.29 in favor of the null (or 4.29 times more likely to occur under the null than the alternative) for Experiment 1, 2.88 in favor of the null for Experiment 2, and 4.09 in favor of the null for Experiment 3. Importantly, for these and all subsequently reported Bayes factors, we also tested a range of priors using the JASP (Version 0.8.1.2; JASP Team, 2017) software's Bayesian one-sample t -test procedure (and one case of a Bayesian paired-samples t -test procedure below). We computed Bayes factors using a range of priors from diffuse to highly informed (informing each of our subsequent experiments on a particular measure with our previous ones), and none ever caused the Bayes factor to flip in favor of the alternative. All still favored the null in all experiments in which there is a null effect reported here.

Although no individual experiment resulted in any detectable overall prediction, a mini-meta-analysis across the three experiments suggested a small but significant overall pattern of above-chance prediction ability, $d = 0.135$, 95% CI = [0.02, 0.25], $Z = 2.32$, $p = .02$. Thus, there may be a slight overall tendency or trend to select the correct direction during retrieval failure, but the effect is small and difficult to detect in an individual experiment.

We turn now to our primary focus: déjà vu states. It is possible that predictive ability during retrieval failure is more likely to occur during déjà vu states than non-déjà vu states. If so, conditionalizing the probability of

correctly choosing the direction on reports of déjà vu should lead to significant ability to predict the direction. However, such conditionalization led to no significant predictive ability in Experiment 1 ($M = .48$, 95% CI = [.41, .56]), $t(68) = -0.49$, $p = .623$, $d = -0.06$, 95% CI = [-0.31, 0.43]; Experiment 2 ($M = .50$, 95% CI = [.45, .55]), $t(92) = -0.02$, $p = .98$, $d = -0.002$, 95% CI = [0.001, 0.004]; or Experiment 3 ($M = .54$, 95% CI = [.50, .59]), $t(106) = 1.77$, $p = .079$, $d = 0.17$, 95% CI = [-0.07, 0.41]. Note that participants who reported no déjà vu states among configurally similar test scenes were necessarily lost from this analysis, as reflected in the degrees of freedom. The BF_{01} favored the null hypothesis in all three experiments: In Experiment 1, it was 6.74 in favor of the null; in Experiment 2, it was 8.72 in favor of the null; and in Experiment 3, it was 2.07 in favor of the null. Furthermore, a meta-analysis across the three experiments indicated no above-chance predictive ability for these conditionalized data, $d = 0.08$, 95% CI = [-0.04, 0.20], $Z = 1.37$, $p = .17$. Thus, there was no indication across all three experiments that déjà vu states were associated with any predictive ability.

Critically, despite lacking in ability to predict in which direction navigation should proceed next when experiencing déjà vu during retrieval failure, participants demonstrated significantly stronger feelings of prediction regarding which direction to turn next during reported déjà vu states ($M = 4.55$, 95% CI = [4.15, 4.94]) than during reported non-déjà vu states ($M = 2.21$, 95% CI = [1.84, 2.57]) in Experiment 2, $t(95) = 13.69$, $p < .001$, $d_z = 1.33$, 95% CI = [1.07, 1.60]. The same pattern of stronger feelings of prediction regarding the direction of the next turn during déjà vu states ($M = 4.42$, 95% CI = [4.08, 4.77]) than during non-déjà vu states ($M = 1.43$, 95% CI = [1.18, 1.70]) occurred in Experiment 3, $t(109) = 16.29$, $p < .001$, $d_z = 1.54$, 95% CI = [1.27, 1.80]. Note that participants who reported no déjà vu states were still lost from this analysis, but that participants who indicated déjà vu states only among test scenes corresponding to nonstudied scenes could now be included (unlike with the probability of correct direction selection when déjà vu was indicated); this is reflected in the degrees of freedom. A meta-analysis of the feeling-of-prediction ratings obtained in Experiments 2 and 3 revealed a large difference between feeling-of-prediction ratings given during reported déjà vu and reported non-déjà vu states ($d = 1.42$, 95% CI = [1.23, 1.62], $Z = 13.98$, $p < .001$). This difference was such that higher feeling-of-prediction ratings were given during reported déjà vu states than non-déjà vu states. Taken together with the lack of predictive ability during reported déjà vu states (see Fig. 2), the full pattern suggests that déjà vu states led to an illusory sense of knowing what would happen next.

Figure 2 depicts the individual participants' data for their feeling-of-prediction ratings during déjà vu versus non-déjà vu reports (left column) and their actual prediction performance during reported déjà vu states (right column), illustrating the illusory nature of the feeling of prediction during déjà vu. Interestingly, participants' feeling-of-prediction ratings were significantly higher for configurally similar ($M = 2.97$, 95% CI = [2.64, 3.32]) than dissimilar scenes ($M = 2.79$, 95% CI = [2.45, 3.14]), in Experiment 2, $t(102) = 2.61$, $p = .010$, $d_z = 0.26$, 95% CI = [0.06, 0.45]. The same pattern emerged in Experiment 3, with higher feeling-of-prediction ratings for configurally similar ($M = 2.45$, 95% CI = [2.15, 2.74]) than dissimilar ($M = 2.22$, 95% CI = [1.95, 2.49]) scenes, $t(120) = 3.63$, $p < .001$, $d_z = 0.34$, 95% CI = [0.15, 0.52]. In both experiments, feeling-of-prediction ratings followed the same pattern as the familiarity ratings, suggesting that the familiarity brought on by configural resemblance may have contributed to the feeling of prediction regarding what would happen next during navigation of the scene. This points toward the possibility that memory-driven factors that contribute to the sense of déjà vu may also contribute to an illusory sense of knowing what will happen next. This is a hypothesis for future research.

Because remembering the direction of the turn taken in a scene was difficult even when scene recall succeeded, a concern might be a low likelihood of detecting an above-chance rate of turn prediction on trials in which recall failed, as there was little room for lower performance. Experiment 3 was set up to encourage greater attention to the navigational path at encoding, and although this manipulation did directionally increase prediction performance during recall success, such performance was still only at .61. In addition, the probability of choosing the correct turn during reported déjà vu states appeared to be approaching significance in Experiment 3. To address this issue further, we examined only those participants whose predictive accuracy during recall success was .70 or greater, and we conducted a meta-analysis on these data across all three experiments. In Experiment 1, 15 participants met this criterion ($M = .81$, 95% CI = [.76, .86]). In Experiment 2, 22 participants met it ($M = .85$, 95% CI = [.80, .89]), and in Experiment 3, 40 participants met it ($M = .82$, 95% CI = [.78, .85]). In none of these experiments did this subset of participants demonstrate above-chance predictive ability (where chance was .50) when reporting déjà vu states during recall failure—Experiment 1: $M = .51$, 95% CI = [.34, .67], $t(14) = 0.15$, $p = .89$, $d = 0.04$, 95% CI = [-0.02, 0.10]; Experiment 2: $M = .50$, 95% CI = [.40, .59], $t(21) = -0.01$, $p = .99$, $d = -0.002$, 95% CI = [0.000, 0.004]; Experiment 3 (not all of these participants reported déjà vu, which explains the degrees

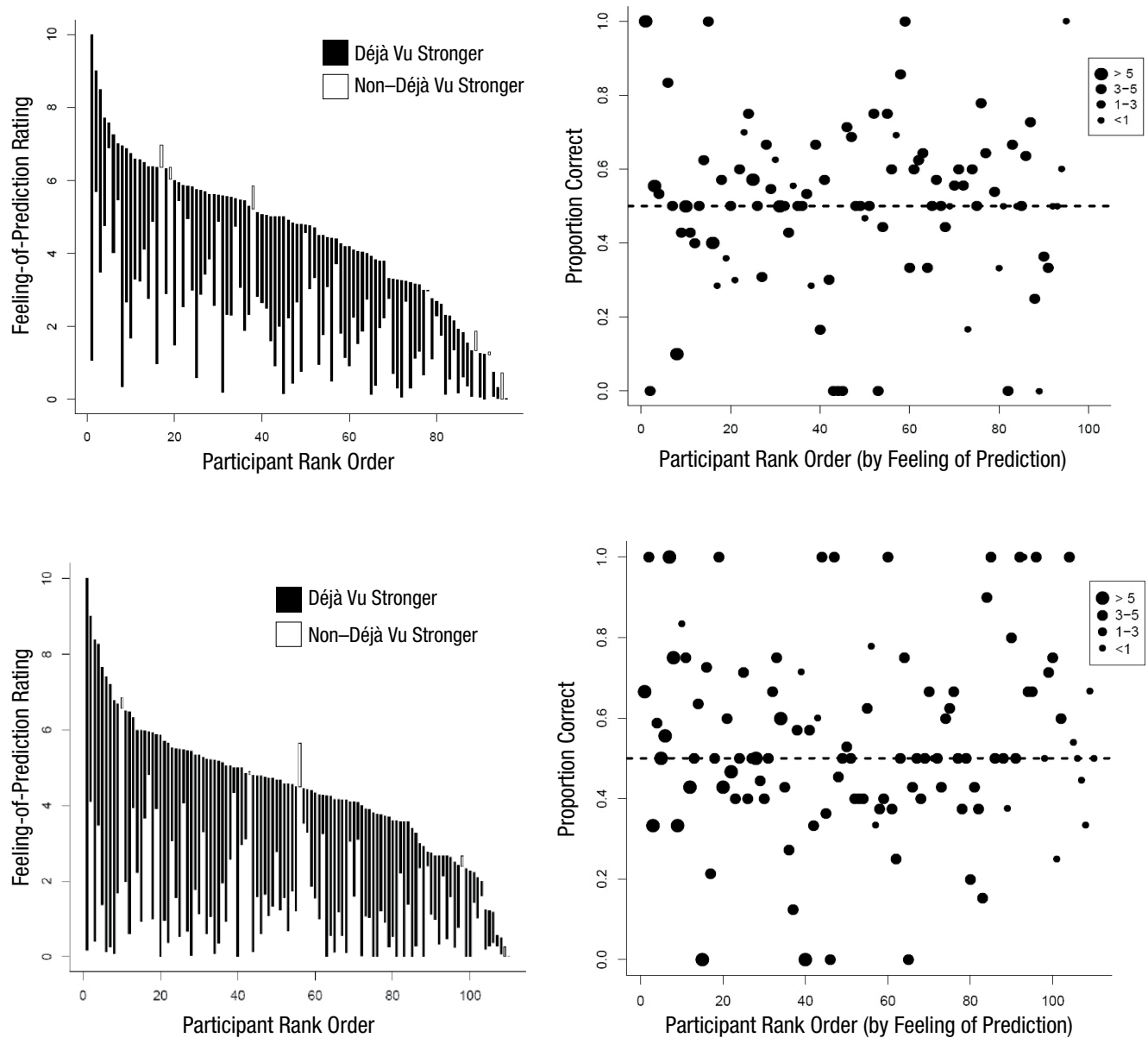


Fig. 2. Individual participant data for feeling-of-prediction ratings (left) and actual prediction performance during déjà vu (right) for Experiment 2 (top row) and Experiment 3 (bottom row). The individual-data plots are rank-ordered by highest to lowest feeling-of-prediction rating given during reported déjà vu states, for comparability among participants from left to right. In the graphs on the left, dark bars indicate instances in which the mean feeling-of-prediction rating was higher during déjà vu than non-déjà vu states. Clear bars indicate instances in which that pattern was reversed. The top end of each dark bar indicates the mean rating given among instances of reported déjà vu; the bottom end indicates the mean rating given among instances of non-déjà vu; and the length of the bar indicates the magnitude of the difference for that person. The same holds true but in the reverse order for the clear bars. In the plots on the right, the dots indicate individuals' accuracy level in choosing the correct direction of the next turn relative to chance (chance = .50). The larger dots represent participants for whom the difference between feeling-of-prediction ratings during déjà vu vs. non-déjà vu states (in the left-hand panel) was greater than 5, with the smaller dots representing participants for whom the magnitude of the déjà vu feeling-of-prediction ratings effect was smaller. As can be seen here, there was no relationship between the strength of the déjà vu feeling-of-prediction effect and actual prediction ability during déjà vu, suggesting that the feeling of prediction during reported déjà vu states was illusory.

of freedom): $M = .55$, 95% CI = [.49, .62], $t(36) = 1.40$, $p = .17$, $d = 0.23$, 95% CI = [−0.24, 0.70]. The BF_{01} favored the null in all three experiments: In Experiment 1, it was 3.77 in favor of the null; in Experiment 2, it was 4.48 in favor of the null; and in Experiment 3, it was 2.31 in favor of the null. A meta-analysis of these

data revealed no significant pattern of above-chance discrimination, $d = 0.12$, 95% CI = [−0.11, 0.34], $Z = 1.02$, $p = .31$.

In further support of the idea that there is no relationship between the ability to remember the direction of the final turn when recall of the scene itself

succeeded and ability to predict the turn during déjà vu states when recall failed, there were no significant correlations between these two variables in any experiment (Experiment 1: $r = .11$, 95% CI = $[-.13, .34]$, $p = .39$; Experiment 2: $r = .04$, 95% CI = $[-.17, .24]$, $p = .74$; Experiment 3: $r = -.03$, 95% CI = $[-.22, .16]$, $p = .74$), nor did a meta-analysis of the correlations across the experiments reveal a pattern, $r = .03$, 95% CI = $[-.10, .147]$, $Z = .42$, $p = .67$.

Finally, some might wonder if it is only the higher feeling-of-prediction ratings that are associated with predictive ability during recall failure; if so, then we should find higher average feeling-of-prediction ratings for correct than incorrect answers. However, feeling-of-prediction ratings were virtually equal for correct ($M = 2.98$, 95% CI = $[2.57, 3.35]$) and incorrect ($M = 2.94$, 95% CI = $[2.59, 3.26]$) left-right responses for Experiment 2, $t(102) = 0.39$, $p = .70$, $d_z = 0.04$, 95% CI = $[-0.16, 0.23]$, as well as for Experiment 3 (correct: $M = 2.45$, 95% CI = $[2.13, 2.73]$; incorrect: $M = 2.46$, 95% CI = $[2.13, 2.81]$), $t(120) = -0.11$, $p = .91$, $d_z = 0.01$, 95% CI = $[-0.18, 0.17]$. The BF_{01} favored the null in both experiments: In Experiment 2, it was 8.51 in favor of the null, and in Experiment 3, it was 9.84 in favor of the null. A meta-analysis across the two experiments also revealed no significant difference, $d = 0.01$, 95% CI = $[-0.12, 0.14]$, $Z = 0.18$, $p = .85$.

In summary, the full pattern of results points toward a stronger feeling of prediction during déjà vu than non-déjà vu even though there is no evidence of actual prediction ability during déjà vu states or of any association between the feelings of prediction and the likelihood of actual prediction. This is not to say that memory-based prediction ability during déjà vu states can never happen—the main take-home message of the present study is that during déjà vu, people report an increased feeling of knowing what will happen next in a situation in which they do not appear to actually know what will happen next.

Discussion

This study presents a significant advance in understanding the peculiar association between déjà vu and feelings of premonition (e.g., Brown, 2004). The results suggest a link between the sensation of déjà vu and a bias to believe that one knows what will happen next. Why exactly the déjà vu bias occurs is the next step for future investigation. However, some clues may be found in the recent discovery that TOT states also lead to bias (Cleary & Claxton, 2015). In the case of the TOT bias, the researchers surmised that because the TOT state is itself a feeling of being on the verge of retrieving a sought-after word, the feeling itself might lead people to infer characteristics consistent with fluency or

accessibility (because being on the verge of retrieving a word might be taken by the experiencer to indicate that the word is in a heightened state of accessibility relative to if no TOT state were present; this, in turn, might be taken to indicate an increased likelihood of accessible characteristics).

In the case of déjà vu, participants might likewise infer from the state itself that a memory is right on the verge of access—if the memory is assumed to be temporally dynamic (as most real-life events are), it might lead to the erroneous inference that how the event unfolds is also right on the verge of access and thus about to readily come to mind. This could lead to a self-perpetuating effect in real-life, whereby because a person felt on the verge of retrieving what would come next, and because what did then happen next continued to feel familiar, familiarity then is misinterpreted as an indication that the person successfully predicted what would come next. Such a self-perpetuating feedback effect of the familiarity feeling could feed into an erroneous perception that déjà vu is often accompanied by not only a sense of knowing what will happen next but also a false confirmatory sense of being correct regarding what actually did come next.

It is important to note that while we have demonstrated that people can be biased during déjà vu states toward thinking that they know what will happen next when they do not, we have not ruled out the possibility that memory-driven déjà vu can, in other situations, drive actual predictive ability. It may be that a bias exists in the first place because sometimes memory-driven feelings of déjà vu do lead to successful prediction in the world. Although our meta-analysis of overall prediction ability during retrieval failure across experiments showed a trend toward a small predictive-ability effect, two factors point toward the possibility that this tiny effect was due to an unconscious or implicit form of memory (e.g., Cleary, 2012; Kunst-Wilson & Zajonc, 1980). First, such predictive ability did not occur when only reported déjà vu states were considered. Second, feelings of prediction were unrelated to correctly or incorrectly predicting the direction of the turn.

Could participants have believed that the experiencers were defining the déjà vu state itself as a feeling of prediction or even have defined it that way themselves? Cleary et al. (2009) found that whether a déjà vu definition was provided did not impact their similarity-manipulations' effects on déjà vu likelihood and, also, that fewer than 14% of participants tended to pre-experimentally define déjà vu as a feeling of premonition. Moreover, participants do not appear to report feelings of prediction every time they experience déjà vu. Although the present study used a rating of the intensity of the feeling of prediction, ongoing work in our laboratory using a yes/no question suggests that

it is roughly half of déjà vu reports that are accompanied by feelings of prediction.

Some readers may note that the rates of déjà vu reports seem higher here and in past studies (e.g., Cleary et al., 2012) than in daily life. Referencing an analogous issue in the TOT literature, whereby TOT rates are higher in the lab than in daily life, Cleary (2014) argues that the laboratory situation in both cases (déjà vu and TOT states) is explicitly designed to favor the occurrence of the phenomenon on the basis of hypotheses about contributing factors.

Finally, an important facet of the present work is that we demonstrated a novel connection between memory and future-orientedness. Other researchers have suggested that memory's primary adaptive purpose is its enabling of future prediction more so than its allowing recollection of the past (Schacter et al., 2007; Szpunar et al., 2014). The present study demonstrates that there is a link between subjective states of memory, such as déjà vu, and thinking about the future. The fact that this link exists points further toward the idea that memory is itself a future-oriented process.

Action Editor

D. Stephen Lindsay served as action editor for this article.

Author Contributions

A. M. Cleary developed the study idea, the study design, and the stimuli. Testing and data collection for Experiments 1 and 2 were performed by A. B. Claxton; both A. M. Cleary and A. B. Claxton collected data for Experiment 3. Most data were analyzed and interpreted by A. M. Cleary; A. B. Claxton served as a second set of eyes on the data coding and also performed effect size and confidence interval calculations around effect sizes. A. M. Cleary drafted the manuscript, and A. B. Claxton provided critical revisions. Both authors approved the final version of the manuscript for submission.

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Declaration of Conflicting Interests

The author(s) declared that there were no conflicts of interest with respect to the authorship or the publication of this article.

Open Practices



All data and materials have been made publicly available via the Open Science Framework (OSF) and can be accessed

at <https://osf.io/62wn9/>. The design and analysis plans for Experiment 3 were preregistered at OSF and can be accessed at <https://osf.io/tqp3d/>. The complete Open Practices Disclosure for this article can be found at <http://journals.sagepub.com/doi/suppl/10.1177/0956797617743018>. This article has received badges for Open Data, Open Materials, and Preregistration. More information about the Open Practices badges can be found at <http://www.psychologicalscience.org/publications/badges>.

Note

1. In one instance (Experiment 1, the probability of reporting déjà vu among test scenes that mapped onto earlier-studied scenes vs. among those that did not), the confidence intervals around d_z as computed using *metafor* did not align with the *t*-test results, prompting a manual calculation for the exact CI as specified in an update posted by Wolfgang Viechtbauer (<https://stats.stackexchange.com/questions/87068/conflict-in-confidence-intervals-for-mean-difference-and-confidence-interval-for>). Because it relies on an asymptotic approximation using a normal distribution, the CI around the effect size calculated in *metafor* can lead to a result that does not align with the results of the *t* test. For this, Viechtbauer recommends manual calculation of the exact CI, with instructions at the above link, which we used for Experiment 1.

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